

The Current Popularity and the Future of Docker

Docker containers enable developers and operators to build and ship an application anywhere. The author discusses the popularity of Docker container technology, and its future relevance and scope.



Virtualisation refers to the creation of a ‘virtual’ technology—an abstraction built to support a diverse set of applications and systems using the same base set of resources. Virtual machines (VMs) are the implementation of virtualisation for an entire machine. They emulate the hardware, networking and storage, providing an isolated system for the end user. A VM provides the complete set of functionality an operating system would have provided. Multiple VMs can be run on a single system, thus optimising the usage of system resources and reducing the need for separate hardware for each system. Specialised software, known as a hypervisor, is often used on servers to manage the different VMs. If you have surfed the Internet, you have probably been exposed to a virtual machine without realising it. That is the beauty of this technology!

The technology has, in fact, developed over a span of decades, starting with IBM’s first commercial mainframe that supported virtualisation, which evolved to address the demand by Bell Labs and the Massachusetts Institute of Technology in the 1960s.

Containers

You’d think containers were a modern update to virtual machines, but the fact is that they date back to the 1970s, to the ‘chroot’ of UNIX machines (a command that defined a new root directory for the system), introducing the concept of isolating processes from each other without the need for hardware-level virtualisation. The Linux container was one of the first comprehensive implementations of the container ecosystem. A series of updates and several